

Carlotta Barone-MacDonald

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EDUCATION

- MIT**—MS, Electrical Engineering & MS, Technology and Policy CAMBRIDGE
- **Relevant Coursework:** Applied Quantum and Statistical Physics, Quantum Engineering Platforms; Quantum Technology and Devices; How to Make (almost) Anything; Machine Learning; ML for Nuclear Science and Engineering
 - **Leadership Positions:** MIT Quantum Hackathon Executive Team, Global Teaching Labs South Africa Quantum Teacher, Technology and Policy Student Society Board Member, MIT Policy Hackathon Quantum Challenge Lead
- École Polytechnique**—BS, Mathematics and Physics Double Major PALAISEAU
- **Relevant Coursework:** Thermodynamics and Statistical Physics; Introduction to Quantum Mechanics; Wave Optics and Radiation; Electrodynamics; Fluid Mechanics; Probability & Statistics; Numerical Optimization
 - **Awards:** Honors Student, Outstanding Student Distinction, Fawco Foundation—Women in STEM Education Award
 - **Leadership Positions:** Music Association Board Member, Debate Committee Board Member, BX-Pro Founding Member
- King's College London**—Visiting Student LONDON
- **Relevant Coursework:** Advanced Quantum Mechanics, Ethics, Mathematical Aspects of Statistical Mechanics

EXPERIENCE

- Massachusetts Institute of Technology** CAMBRIDGE, 2024
- Advisors: Dr. Florian Metzler, Prof. Peter Hagelstein and Prof. Paola Cappellaro
- Graduate Researcher, Master's Thesis** 2025—Present
- Analyzing the development of emerging quantum technologies by constructing quantitative and qualitative indicators of scientific credibility, funding dynamics, and field evolution across national and institutional environments.
- Graduate Research Assistant** 2024—Present
- Developing an experimental platform to investigate collective nuclear effects in gas-loaded metal–hydrogen systems.
 - Design and assembly of vacuum and gas-loading setups, including component selection, system integration, and leak testing.
 - Developing a CR-39 charged-particle track-analysis pipeline in Python, integrating *ilastik*-based preprocessing, feature extraction, and ML-based classification to distinguish track morphologies.
 - Calibrated liquid scintillation neutron detectors (Eljen) using Cf-252, and established reproducible background baselines.
- Undergraduate Researcher, Undergraduate Thesis** 2024
- Studied delocalized quantum states in quantum annealing and exciton diffusion, connecting energy-transport and information-transport mechanisms.
 - Developed and simulated models for exciton diffusion using Heisenberg Hamiltonians and quantum random walks.
 - Investigated quantum annealing for optimization, implementing a shortest-path solver using QuTiP in Python.
- Imperial College London, Controlled Quantum Dynamics Group** LONDON, 2023
- Undergraduate Researcher** under Prof. Myunshik Kim and Soovin Lee
- Investigated methods and algorithms for computing stabilizer entropies in quantum systems.
 - Implemented QFT and Grover's search in QuTiP and analyzed the evolution of nonstabilizerness during circuit execution.
- SES-Imagotag, Environmental, Social, and Governance team** PARIS, 2023
- Summer Mission Intern** under Pascale Dubreuil
- Collaborated with Carbone4 consultants to model carbon-emissions reductions enabled by SES-Imagotag products.

PROJECTS & PUBLICATIONS

- TrackTomic: Machine Learning for CR-39 Nuclear Track Analysis** 2025
- Developing a machine learning pipeline to infer latent physical information, such as particle energy and direction, from etched CR-39 nuclear track detector images. Integrating unsupervised and physics-informed learning approaches to characterize etch pit geometries and infer source parameters, with potential applications in neutron and charged particle detection.
- Wearable Audio Device (HTMAA Final Project)** 2025
- Designing a wearable audio device with a XIAO ESP32-S3, I2S amplifier, capacitive-touch interface, and custom PCB; implemented firmware, power management, and a 3D-printed enclosure.

SKILLS & SELECTED AWARDS

COMPUTATIONAL	Python (QuTip, Numpy, Matplotlib, Jupyter, SciPy, PyTorch), R, $\LaTeX$, Git
HARDWARE & FABRICATION	CAD, PCB Design, Electronics Assembly, Embedded Programming, 3D Printing, Laser Cutting
TRANSVERSAL	Data Analysis, Experimentation, Leadership, Public Speaking & Critical Thinking
LANGUAGES	French (Native), English (Native), Spanish (Fluent), Italian (A2)
SELECTED AWARDS	French Female Mathematics Olympiads—1 st place (2021); American Legion—Gold Award Scout of the Year (2022); Fawco Foundation—Award for women in STEM education (2022)